5.3 Local Extrema and Concavity

Name _____ Period ____ Date _____

AP Calculus 5.3 (30-minute plan) · Student sheet

I can…

- use the First Derivative Test to find local extrema.
- find where a graph is concave up or down and find inflection points.
- use the Second Derivative Test.

Vocabulary · fill in the missing word

First Derivative Test — if f′ _____ from + to − at c, f(c) is a local max; from − to +, a local min

Concave up — f″ > 0: the graph bends _____ (its slopes increase)

Inflection point — a point on the graph where the _____ changes

Second Derivative Test — f′(c) = 0 and f″(c) < 0: _____ max; f″(c) > 0: local min

Part 1 · Notes and examples (fill in each note, then work the example)

First Derivative Test: if f′ changes from positive to negative at c, then f(c) is a local _____.

Example 1

Find the local extrema of $f(x) = x^3 - 3x^2 - 9x + 1$.

Concavity: if f″(x) > 0 on an interval, the graph is concave _____ there.

Example 2

Where is $f(x) = x^3 - 3x^2 - 9x + 1$ concave up? Concave down?

Inflection point: a point where the _____ changes.

Second Derivative Test: if f′(c) = 0 and f″(c) < 0, then f(c) is a local _____.

Example 3

Find the inflection point of $f(x) = x^3 - 3x^2 - 9x + 1$.

Part 2 · Partner practice

Partner practice

(AP® Practice) The graph of $f(x) = x^3 - 6x^2 + 5$ has an inflection point at $x =$

(A) 2 (B) 0 (C) 4 (D) 6